

E-commerce Customer Segmentation & Retention

Olist Brazilian marketplace — 96,478 delivered orders, 93,358 unique customers, 2016-09-15 to 2018-08-29

93,358	3.0%	R\$160	R\$15.4M
Unique customers	Repeat purchase rate	Avg order value	Total revenue

So what: 30% of customers (Champions + At-Risk High Value) generate 54% of revenue. The At-Risk segment alone represents ~R\$4.1M in past spend that has gone cold — the highest-leverage move is a win-back campaign, not a loyalty program for a repeat-buyer base that barely exists.

Key Findings

- Repeat purchase rate is only 3.0% — 97 of 100 customers never buy again. This is an acquisition-funded marketplace, not a retention-funded one.
- Month-1 retention averages 5.5% across acquisition cohorts, with no improvement over the 2-year window — churn is structural, not seasonal.
- At-Risk (high value, lapsed) customers: 14.6% of customers, 26.6% of revenue, avg R\$302/customer — proven spenders who have gone quiet.
- Champions: 15.3% of customers, 27.2% of revenue — the base that's already working; protect it, don't over-invest in acquiring more like it at the expense of win-back.
- Data quality catch: Olist's customer_id is a per-order surrogate key, not a person identifier — RFM was built on customer_unique_id to avoid a false 0% repeat-rate read.

Charts

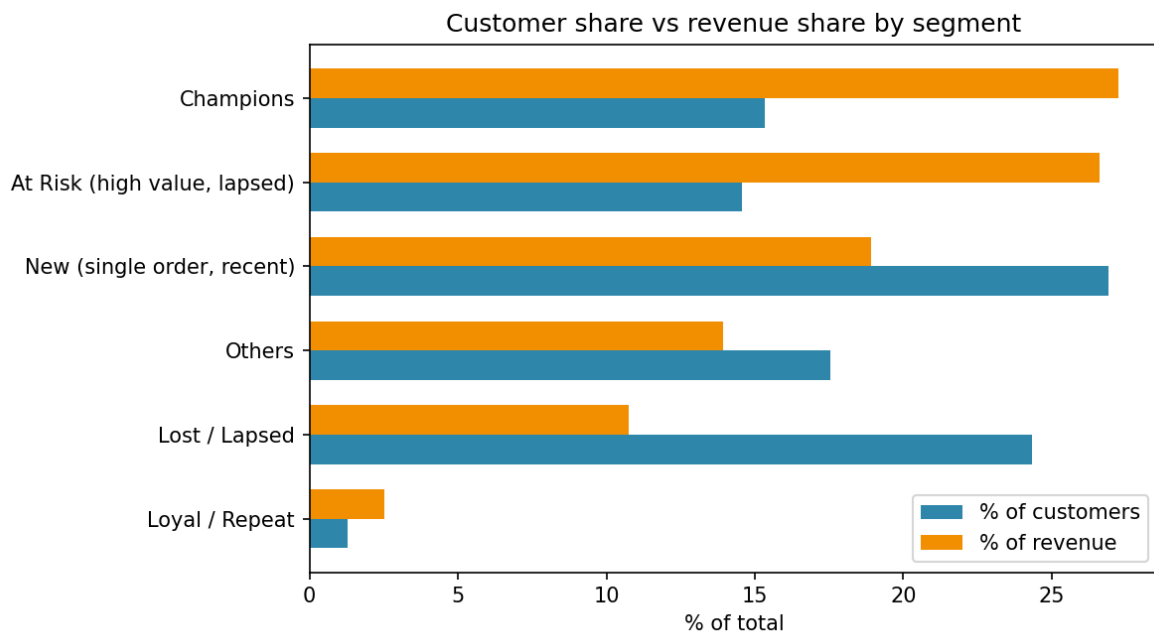


Fig 1. Customer share vs. revenue share by segment — the gap between bars is the concentration story.

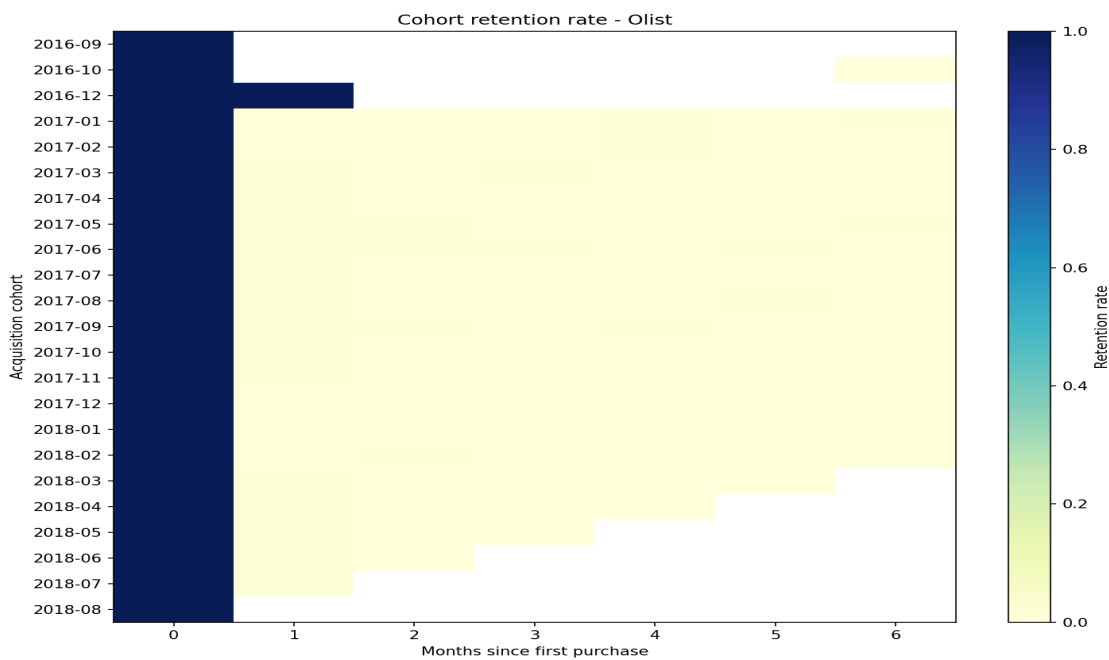


Fig 2. Cohort retention heatmap — every acquisition cohort fades to near-zero within 1-2 months.

Method

- Filtered to `order_status == 'delivered'` to exclude cancelled/failed transactions.
- Order value = sum of `payment_value` per order (captures installments/vouchers correctly).
- RFM computed per `customer_unique_id` (not `customer_id`), scored into quintiles, mapped to 6 business segments.
- Cohort retention: customers grouped by first-purchase month, tracked forward by months-since-acquisition.

Recommendation

- Build a win-back flow (targeted discount/email) for the At-Risk, high-value segment — clear ROI given proven spend history.
- Investigate why month-1 retention doesn't recover post-purchase — likely delivery experience or marketplace-seller trust given Olist's multi-seller model; cut by review_score as a follow-up.
- Treat 'New (single order, recent)' (27% of customers) as the real growth lever — a small lift in second-order conversion has outsized impact given how few customers reach that stage today.

Source: Olist Brazilian E-Commerce Public Dataset (Kaggle). Full pipeline reproducible via [analysis.py](#); interactive version in [dashboard.html](#).